

Physics 514, Problem Set 3: Geodesics and Covariant Derivatives

due: *Tuesday, February 8, 5 PM*

Please place your solutions in the box outside my office (Rutherford 316) before the due date. You are encouraged to discuss these problems with your colleagues, but you must write up your own solutions; the solutions you hand in should reflect your own work and understanding.

Reading: Sections 3.1-3.4 of Carroll.

1. In class we derived the geodesic equations

$$\ddot{x}^\mu + \Gamma_{\nu\rho}^\mu \dot{x}^\nu \dot{x}^\rho = f(\lambda) \dot{x}^\mu$$

by varying the action

$$S[x^\mu(\lambda)] = \int \sqrt{g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu} d\lambda$$

with respect to $x^\mu(\lambda)$. Show that if λ is an affine parameter (so that $f(\lambda) = 0$) one can also obtain the geodesic equations by varying the action

$$S'[x^\mu(\lambda)] = \int g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu d\lambda$$

This action is *much* simpler than the previous action. So in practice a very fast way of computing Christoffel symbols is to write down the Euler-Lagrange equations for $S'[x^\mu(\lambda)]$ and then read off the Christoffel symbols from the resulting geodesic equation. You may find it easier to use this method to compute Christoffel symbols in the future.

2. Chapter 3, Problem 3.
3. Chapter 3, Problem 4. For parts (d) and (e) please compute the Christoffel symbols in parabolic coordinates using the formula for the metric you found in part (c). Use this result to compute the derivatives in part (e) in parabolic coordinates.
4. Chapter 3, Problem 5a. (You may do part (b) if you like, but it is not required)
5. Chapter 3, Problem 6.

1.

$$S[x^\mu(\lambda)] = \int g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu d\lambda$$

Using the E-L eqns,

$$\frac{d}{d\lambda} \left(\frac{\partial}{\partial \dot{x}^\alpha} (g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu) \right) = \frac{\partial}{\partial x^\alpha} (g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu)$$

$$\Rightarrow \frac{d}{d\lambda} (g_{\mu\nu} (\delta_\alpha^\mu \dot{x}^\nu + \delta_\alpha^\nu \dot{x}^\mu)) = \dot{x}^\mu \dot{x}^\nu \partial_\alpha g_{\mu\nu}$$

$$\Rightarrow \frac{d}{d\lambda} (2g_{\alpha\mu} \dot{x}^\mu) = \dot{x}^\mu \dot{x}^\nu \partial_\alpha g_{\mu\nu}$$

$$\Rightarrow 2(g_{\alpha\mu} \ddot{x}^\mu + \dot{x}^\mu \dot{x}^\nu \partial_\nu g_{\alpha\mu}) = \dot{x}^\mu \dot{x}^\nu \partial_\alpha g_{\mu\nu}$$

$$\Rightarrow g^{\rho\alpha} g_{\alpha\mu} \ddot{x}^\mu + \dot{x}^\mu \dot{x}^\nu g^{\rho\alpha} \partial_\nu g_{\alpha\mu} \stackrel{= \dot{x}^\nu \dot{x}^\mu g^{\rho\alpha} \partial_\mu g_{\nu\alpha}}{\rightarrow} -\frac{1}{2} \dot{x}^\mu \dot{x}^\nu g^{\rho\alpha} \partial_\alpha g_{\mu\nu} = 0$$

$$\Rightarrow \ddot{x}^\rho + \frac{1}{2} g^{\rho\alpha} (\partial_\nu g_{\alpha\mu} + \partial_\mu g_{\nu\alpha} - \partial_\alpha g_{\mu\nu}) \dot{x}^\mu \dot{x}^\nu = 0$$

$$\Rightarrow \ddot{x}^\rho + \Gamma_{\mu\nu}^\rho \dot{x}^\mu \dot{x}^\nu = 0.$$

This is indeed the geodesic eqn. with $f(\lambda) = 0$.

2. Carroll 3.3

$$\frac{1}{\sqrt{|g_{\lambda\lambda}|}} \cdot \frac{1}{2} (|g_{\lambda\lambda}|)^{-1/2} \partial_\mu |g_{\lambda\lambda}|$$

3. Imagine we have a *diagonal* metric $g_{\mu\nu}$. Show that the Christoffel symbols are given by

$$\Gamma_{\mu\nu}^\lambda = 0 \quad (3.213)$$

$$\Gamma_{\mu\mu}^\lambda = -\frac{1}{2} (g_{\lambda\lambda})^{-1} \partial_\lambda g_{\mu\mu} \quad (3.214)$$

$$\Gamma_{\mu\lambda}^\lambda = \partial_\mu (\ln \sqrt{|g_{\lambda\lambda}|}) \quad (3.215)$$

$$\Gamma_{\lambda\lambda}^\lambda = \partial_\lambda (\ln \sqrt{|g_{\lambda\lambda}|}) \quad (3.216)$$

In these expressions, $\mu \neq \nu \neq \lambda$, and repeated indices are *not* summed over.

$$\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\rho} (\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu})$$

Suppose $g_{\mu\nu}$ (and thus $g^{\mu\nu}$) is diagonal. Also assume $\mu \neq \nu \neq \lambda$.

For this problem, indices only represent numbers & I will be lax with upper/lower indices.

$$\text{Then, } \Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\lambda} (\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu})$$

since all $\rho \neq \lambda$ vanish due to $g^{\rho\lambda}$.

$$= 0 \quad \text{since } \nu \neq \mu \neq \lambda, \text{ \& } g \text{ is diagonal.}$$

$$\begin{aligned} \text{Next, } \Gamma_{\mu\mu}^\lambda &= \frac{1}{2} g^{\lambda\lambda} (\partial_\mu g_{\mu\lambda} + \partial_\mu g_{\lambda\mu} - \partial_\lambda g_{\mu\mu}) \\ &= -\frac{1}{2} (g_{\lambda\lambda})^{-1} \partial_\lambda g_{\mu\mu} \end{aligned}$$

where I've used $g^{\lambda\lambda} = (g_{\lambda\lambda})^{-1}$, which holds for diagonal matrices.

Moving on,

$$\Gamma_{\mu\lambda}^{\lambda} = \frac{1}{2} g^{\lambda\lambda} (\partial_{\mu} g_{\lambda\lambda} + \partial_{\lambda} g_{\lambda\mu} - \partial_{\lambda} g_{\mu\lambda})$$

$$= \frac{1}{2 g_{\lambda\lambda}} \partial_{\mu} g_{\lambda\lambda}$$

$$= \frac{1}{2 |g_{\lambda\lambda}|} \partial_{\mu} |g_{\lambda\lambda}|$$

$$= \left\{ \frac{1}{2 \sqrt{|g_{\lambda\lambda}|}} \right\} \left\{ \frac{1}{\sqrt{|g_{\lambda\lambda}|}} \right\} \partial_{\mu} |g_{\lambda\lambda}|$$

$$= \partial_{\mu} \ln \sqrt{|g_{\lambda\lambda}|}$$

And finally $\Gamma_{\lambda\lambda}^{\lambda} = \partial_{\lambda} \ln \sqrt{|g_{\lambda\lambda}|}$ follows trivially.

3. Carroll 3.4

4. In Euclidean three-space, we can define paraboloidal coordinates (u, v, ϕ) via

$$x = uv \cos \phi \quad y = uv \sin \phi \quad z = \frac{1}{2}(u^2 - v^2).$$

- Find the coordinate transformation matrix between paraboloidal and Cartesian coordinates $\partial x^\alpha / \partial x^{\beta'}$ and the inverse transformation. Are there any singular points in the map?
- Find the basis vectors and basis one-forms in terms of Cartesian basis vectors and forms.
- Find the metric and inverse metric in paraboloidal coordinates.
- Calculate the Christoffel symbols.
- Calculate the divergence $\nabla_\mu V^\mu$ and Laplacian $\nabla_\mu \nabla^\mu f$.

a. The transformation matrix is:

$$\frac{\partial x^\alpha}{\partial x^{\beta'}} = \begin{pmatrix} v \cos \phi & u \cos \phi & -uv \sin \phi \\ v \sin \phi & u \sin \phi & uv \cos \phi \\ u & -v & 0 \end{pmatrix}$$

Using the obvious way we would find:

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$v = \sqrt{\sqrt{x^2 + y^2 + z^2} - z}$$

$$u = \sqrt{\sqrt{x^2 + y^2 + z^2} + z}$$

So that:

$$\frac{\partial x^\alpha}{\partial x^{\beta'}} = \begin{pmatrix} \frac{x}{2r\sqrt{r+z}} & \frac{y}{2r\sqrt{r+z}} & \frac{z+r}{2r\sqrt{r+z}} \\ \frac{x}{2r\sqrt{r-z}} & \frac{y}{2r\sqrt{r-z}} & \frac{z-r}{2r\sqrt{r-z}} \\ -\frac{y}{x^2+y^2} & \frac{x}{x^2+y^2} & 0 \end{pmatrix}$$

$$r^2 = x^2 + y^2 + z^2$$

There's problems whenever $r = \pm z, 0$.

b. The usual relations give:

$$\partial_{\mu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \partial_\mu$$

$$dx^{\mu'} = \frac{\partial x^{\mu'}}{\partial x^\mu} dx^\mu$$

$$\text{So: } \begin{pmatrix} \partial_u \\ \partial_v \\ \partial_\phi \end{pmatrix} = \begin{pmatrix} v \cos \phi \partial_x + v \sin \phi \partial_y + u \partial_z \\ u \cos \phi \partial_x + u \sin \phi \partial_y - v \partial_z \\ -uv \sin \phi \partial_x + uv \cos \phi \partial_y \end{pmatrix}$$

$$\begin{pmatrix} du \\ dv \\ d\phi \end{pmatrix} = \begin{pmatrix} \frac{x}{2r\sqrt{r+z}} dx + \frac{y}{2r\sqrt{r+z}} dy + \frac{z+r}{2r\sqrt{r+z}} dz \\ \frac{x}{2r\sqrt{r-z}} dx + \frac{y}{2r\sqrt{r-z}} dy + \frac{z-r}{2r\sqrt{r-z}} dz \\ -\frac{y}{x^2+y^2} dx + \frac{x}{x^2+y^2} dy \end{pmatrix}$$

c. $ds^2 = (dx)^2 + (dy)^2 + (dz)^2$

Note that:

$$dx^\mu = \frac{\partial x^\mu}{\partial x^{\mu'}} dx^{\mu'}$$

$$\Rightarrow \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix} = \begin{pmatrix} v \cos \phi \, du + u \cos \phi \, dv - uv \sin \phi \, d\phi \\ v \sin \phi \, du + u \sin \phi \, dv + uv \cos \phi \, d\phi \\ u \, du - v \, dv \end{pmatrix}$$

So, $ds^2 = \underline{v^2 \cos^2 \phi (du)^2} + \underline{uv \cos^2 \phi \, du \, dv}$

~~$- uv^2 \sin \phi \cos \phi \, du \, d\phi$~~ + $\underline{uv \cos^2 \phi \, dv \, du}$ + $\underline{u^2 \cos^2 \phi (dv)^2}$

~~$- u^2 v \sin \phi \cos \phi \, dv \, d\phi$~~ - ~~$uv^2 \sin \phi \cos \phi \, d\phi \, du$~~

~~$- u^2 v \sin \phi \cos \phi \, d\phi \, dv$~~ + $\underline{u^2 v^2 \sin^2 \phi (d\phi)^2}$

+ $\underline{v^2 \sin^2 \phi (du)^2}$ + $\underline{uv \sin^2 \phi \, du \, dv}$ + ~~$uv^2 \sin \phi \cos \phi \, du \, d\phi$~~

+ $\underline{uv \sin^2 \phi \, dv \, du}$ + $\underline{u^2 \sin^2 \phi (dv)^2}$ + ~~$u^2 v \sin \phi \cos \phi \, dv \, d\phi$~~

+ ~~$uv^2 \sin \phi \cos \phi \, d\phi \, du$~~ + ~~$u^2 v \sin \phi \cos \phi \, d\phi \, dv$~~

+ $\underline{u^2 v^2 \cos^2 \phi (d\phi)^2}$ + $u^2 (du)^2 - uv \, du \, dv - uv \, dv \, du$

+ $v^2 (dv)^2$

$$\begin{aligned}
 &= v^2(du)^2 + uv\,du\,dv + uv\,dv\,du + u^2(dv)^2 \\
 &\quad + u^2v^2(d\phi)^2 + u^2(du)^2 - uv\,du\,dv - uv\,dv\,du \\
 &\quad + v^2(dv)^2
 \end{aligned}$$

$$= (v^2+u^2)(du)^2 + (v^2+u^2)(dv)^2 + u^2v^2(d\phi)^2$$

It follows that:

$$g_{\mu\nu} = \begin{pmatrix} v^2+u^2 & 0 & 0 \\ 0 & v^2+u^2 & 0 \\ 0 & 0 & u^2v^2 \end{pmatrix}$$

and

$$g^{\mu\nu} = \begin{pmatrix} \frac{1}{u^2+v^2} & 0 & 0 \\ 0 & \frac{1}{u^2+v^2} & 0 \\ 0 & 0 & \frac{1}{u^2v^2} \end{pmatrix}$$

$$d. \quad \Gamma_{\mu\nu}^{\lambda} = \frac{1}{2} g^{\lambda\rho} (\partial_{\mu} g_{\nu\rho} + \partial_{\nu} g_{\rho\mu} - \partial_{\rho} g_{\mu\nu})$$

$$\Gamma_{\mu\nu}^0 = \frac{1}{u^2 + v^2} \begin{pmatrix} u & v & 0 \\ v & -u & 0 \\ 0 & 0 & -uv^2 \end{pmatrix}$$

$$\Gamma_{\mu\nu}^1 = \frac{1}{u^2 + v^2} \begin{pmatrix} -v & u & 0 \\ u & v & 0 \\ 0 & 0 & -vu^2 \end{pmatrix}$$

$$\Gamma_{\mu\nu}^2 = \begin{pmatrix} 0 & 0 & -1/3 - 1/3 \\ 0 & 0 & 1/3 - 1/3 \\ 1/u & 1/v & 0 \end{pmatrix}$$

$$e. \quad \nabla_{\mu} V^{\mu} = \partial_{\mu} V^{\mu} + \Gamma_{\mu\lambda}^{\mu} V^{\lambda}$$

$$\begin{aligned}
&= \partial_\mu V^\mu + \frac{1}{u^2+v^2} (u \ v \ 0) \begin{pmatrix} V^1 \\ V^2 \\ 0 \end{pmatrix} \\
&\quad + \frac{1}{u^2+v^2} (u \ v \ 0) \begin{pmatrix} V^1 \\ V^2 \\ 0 \end{pmatrix} \\
&\quad + \frac{1}{uv} (v \ u \ 0) \begin{pmatrix} V^1 \\ V^2 \\ 0 \end{pmatrix}
\end{aligned}$$

$$\nabla_\mu \nabla^\mu f = g^{\mu\nu} \nabla_\mu \nabla_\nu f$$

$$= \frac{1}{u^2+v^2} \nabla_u \partial_u f + \frac{1}{u^2+v^2} \nabla_v \partial_v f + \frac{1}{u^2 v^2} \nabla_\phi \partial_\phi f$$

$$\begin{aligned}
&= \frac{1}{u^2+v^2} (\partial_u^2 f + \Gamma_{uu}^2 \partial_2 f) + \frac{1}{u^2+v^2} (\partial_v^2 f + \Gamma_{vv}^2 \partial_2 f) \\
&\quad + \frac{1}{u^2 v^2} (\partial_\phi^2 f + \Gamma_{\phi\phi}^2 \partial_2 f)
\end{aligned}$$

$$\begin{aligned}
&= \frac{1}{u^2+v^2} \partial_u^2 f + \frac{1}{u^2+v^2} \partial_v^2 f + \frac{1}{u^2 v^2} \partial_\phi^2 f \\
&\quad - \frac{1}{u^2+v^2} \left(\frac{1}{u} \partial_u f + \frac{1}{v} \partial_v f \right)
\end{aligned}$$

4. Carroll 3.5(a)

5. Consider a 2-sphere with coordinates (θ, ϕ) and metric

$$ds^2 = d\theta^2 + \sin^2 \theta d\phi^2. \quad (3.217)$$

- (a) Show that lines of constant longitude ($\phi = \text{constant}$) are geodesics, and that the only line of constant latitude ($\theta = \text{constant}$) that is a geodesic is the equator ($\theta = \pi/2$).
- (b) Take a vector with components $V^\mu = (1, 0)$ and parallel-transport it once around a circle of constant latitude. What are the components of the resulting vector, as a function of θ ?

A geodesic $x^\mu(\lambda)$ obeys:

$$\ddot{x}^\mu + \Gamma_{\nu\rho}^\mu \dot{x}^\nu \dot{x}^\rho = 0$$

where $\Gamma_{\nu\rho}^\mu = \frac{1}{2} g^{\mu\alpha} (\partial_\nu g_{\rho\alpha} + \partial_\rho g_{\alpha\nu} - \partial_\alpha g_{\nu\rho})$

We can read off:

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{pmatrix}$$

$$g^{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & \csc^2 \theta \end{pmatrix}$$

So then:

$$\Gamma_{\nu\rho}^0 = \begin{pmatrix} 0 & 0 \\ 0 & -\sin\theta\cos\theta \end{pmatrix}$$

$$\Gamma_{\nu\rho}^1 = \begin{pmatrix} 0 & \cot\theta \\ \cot\theta & 0 \end{pmatrix}$$

Consider a longitude:

$$\begin{pmatrix} x^0 \\ x^1 \end{pmatrix} = \begin{pmatrix} \lambda \\ c \end{pmatrix}$$

Then $\dot{x}^\mu = (1, 0)$ & $\ddot{x}^\mu = 0$. Thus:

$$\begin{aligned} \ddot{x}^0 + \Gamma_{00}^0 \dot{x}^0 \dot{x}^0 + \Gamma_{01}^0 \dot{x}^0 \dot{x}^1 + \Gamma_{10}^0 \dot{x}^1 \dot{x}^0 \\ + \Gamma_{11}^0 \dot{x}^1 \dot{x}^1 = 0, \end{aligned}$$

$$\begin{aligned} \ddot{x}^1 + \Gamma_{00}^1 \dot{x}^0 \dot{x}^0 + \Gamma_{01}^1 \dot{x}^0 \dot{x}^1 + \Gamma_{10}^1 \dot{x}^1 \dot{x}^0 \\ + \Gamma_{11}^1 \dot{x}^1 \dot{x}^1 = 0, \end{aligned}$$

And so longitudes are geodesics.

Next, let $x^\mu(\lambda) = (c, \lambda)$

Then $\dot{x}^\mu = (0, 1)$ & $\ddot{x}^\mu = 0$. Thus:

$$\ddot{x}^0 + \Gamma_{00}^0 \dot{x}^0 \dot{x}^0 + \Gamma_{01}^0 \dot{x}^0 \dot{x}^1 + \Gamma_{10}^0 \dot{x}^1 \dot{x}^0 + \Gamma_{11}^0 \dot{x}^1 \dot{x}^1 = -\sin\theta \cos\theta$$

$$\ddot{x}^1 + \Gamma_{00}^1 \dot{x}^0 \dot{x}^0 + \Gamma_{01}^1 \dot{x}^0 \dot{x}^1 + \Gamma_{10}^1 \dot{x}^1 \dot{x}^0 + \Gamma_{11}^1 \dot{x}^1 \dot{x}^1 = 0.$$

So latitudes are geodesics only when:

$$-\sin\theta \cos\theta = 0$$

$$\Rightarrow \theta = 0, \pi/2, \pi$$

But $0, \pi$ are singularities, so they are excluded.
So the latitude is a geodesic only when $\theta = \pi/2$,
which is the equator.

5.

Carroll 3.6

6. A good approximation to the metric outside the surface of the Earth is provided by

$$ds^2 = -(1 + 2\Phi)dt^2 + (1 - 2\Phi)dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2), \quad (3.218)$$

where

$$\Phi = -\frac{GM}{r} \quad (3.219)$$

may be thought of as the familiar Newtonian gravitational potential. Here G is Newton's constant and M is the mass of the earth. For this problem Φ may be assumed to be small.

- (a) Imagine a clock on the surface of the Earth at distance R_1 from the Earth's center, and another clock on a tall building at distance R_2 from the Earth's center. Calculate the time elapsed on each clock as a function of the coordinate time t . Which clock moves faster?
- (b) Solve for a geodesic corresponding to a circular orbit around the equator of the Earth ($\theta = \pi/2$). What is $d\phi/dt$?
- (c) How much proper time elapses while a satellite at radius R_1 (skimming along the surface of the earth, neglecting air resistance) completes one orbit? You can work to first order in Φ if you like. Plug in the actual numbers for the radius of the Earth and so on (don't forget to restore the speed of light) to get an answer in seconds. How does this number compare to the proper time elapsed on the clock stationary on the surface?

a. Assuming the clock is stationary, $d\theta, d\phi, dr$ vanish. So:

$$\begin{aligned} 1 &= -ds^2(\dot{x}, \dot{x}) = (1 + 2\Phi) dt(\dot{x}) \otimes dt(\dot{x}) \\ &= (1 + 2\Phi) \left(\frac{dt}{dt} \right)^2 \end{aligned}$$

$$\Rightarrow \tau = \sqrt{1 + 2\Phi} t + \tau_0$$

Evidently, farther clocks run faster.

b. Here we let $dr = d\theta = 0$, $r = R$, $\theta = \pi/2$.
But first, let's get all the geodesic eqn's.

I'll use the variational technique.

$$\begin{aligned}
I &= \frac{1}{2} \int g_{\mu\nu} d\dot{x}^\mu d\dot{x}^\nu d\tau \\
&= \frac{1}{2} \int - (1+2\Phi) \left(\frac{dt}{d\tau} \right)^2 + (1-2\Phi) \left(\frac{dr}{d\tau} \right)^2 \\
&\quad + r^2 \left(\frac{d\theta}{d\tau} \right)^2 + r^2 \sin^2 \theta \left(\frac{d\phi}{d\tau} \right)^2 d\tau
\end{aligned}$$

Let $t \rightarrow t + \delta t$. Then it is clear:

$$\begin{aligned}
\delta I &= \frac{1}{2} \int - (1+2\Phi) \left(\frac{d(t+\delta t)}{d\tau} \right)^2 + (1+2\Phi) \left(\frac{dt}{d\tau} \right)^2 d\tau \\
&= \frac{1}{2} \int - (1+2\Phi) \left(2 \frac{dt}{d\tau} \frac{d\delta t}{d\tau} \right) d\tau \\
&= - \int (1+2\Phi) \frac{d^2 t}{d\tau^2} \delta t d\tau
\end{aligned}$$

Then as usual,

(unless $r = 2GM$)

$$(1+2\Phi) \frac{d^2 t}{d\tau^2} = 0 \Rightarrow \frac{dt}{d\tau} = \alpha$$

Similarly, it's easy to see: (if we varied ϕ)

$$\frac{d\phi}{d\tau} = \beta$$

(or $\theta = 0$ or π)

Next let $\theta \rightarrow \theta + \delta\theta$.

Note that $\sin(\theta + \delta\theta) \approx \sin\theta + \delta\theta \cos\theta$

$$\Rightarrow \sin^2(\theta + \delta\theta) \approx \sin^2\theta + 2\delta\theta \sin\theta \cos\theta$$

So then:

$$\delta I = \int \left(r^2 \frac{d^2\theta}{dt^2} + \sin\theta \cos\theta \right) \delta\theta dt$$

This is useless. Next comes $r \rightarrow r + \delta r$.

Then $r^2 \rightarrow r^2 + 2r\delta r$, $\frac{1}{r} \rightarrow \frac{1}{r} - \frac{1}{r^2}\delta r$. So:

$$\delta I = \int \left[-\frac{GM}{r^2} \left(\frac{dt}{dt} \right)^2 - \frac{GM}{r^2} \left(\frac{dr}{dt} \right)^2 + \left(1 - 2\frac{GM}{r} \right) \frac{d^2r}{dt^2} + r \left(\frac{d\theta}{dt} \right)^2 + r \sin^2\theta \left(\frac{d\phi}{dt} \right)^2 \right] \delta r dt$$

Letting $d\theta = dr = 0$, $r = R$, $\theta = \pi/2$, we see:

$$\frac{d\phi}{dt} = \sqrt{\frac{GM}{R^3}}$$

c. Like before,

$$1 = -ds^2(\dot{x}, \dot{x}) = \left(1 - 2\frac{GM}{R_1} \right) \left(\frac{dt}{dt} \right)^2 - R_1^2 \left(\frac{d\phi}{dt} \right)^2$$

$$\Rightarrow \left(\frac{d\tau}{dt}\right)^2 = 1 - \frac{2GM}{R_1} - R_1^2 \left(\frac{d\phi}{dt}\right)^2$$

$$\Rightarrow \frac{d\tau}{dt} = \sqrt{1 + 3\Phi}$$

$$\Rightarrow \tau = \sqrt{1 + 3\Phi} t + \tau_0$$

Now, for one orbit, $\frac{2\pi}{\dot{\phi}} = 2\pi \sqrt{\frac{R^3}{GM}}$ time elapses.

So the proper time elapsed is:

$$\begin{aligned} \tau &= 2\pi \sqrt{1 + 3\Phi} \sqrt{\frac{R^3}{GM}} \\ &= 2\pi R \sqrt{\frac{R}{GM} - 3} \end{aligned}$$

Or, including c 's,

$$\begin{aligned} \tau &= 2\pi R \sqrt{\frac{R}{GM} - \frac{3}{c^2}} \\ &\approx 5069 \text{ s} \end{aligned}$$

If stationary, $3 \rightarrow 2$ & $\tau \approx 5069 \text{ s}$ still.